pfi

# Python for Fixed Income

COURSE 6 · NEURAL NETWORKS

## One-Dimensional Convolutions and Many Tenors

WEEK 3 · DAY 3

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# Three ways to read a window

### Dense, day 11

Every day its own weight: the window laid flat.

### Recurrent, day 12

Day by day, a state carried forward.

### Convolution, today

**Five weights slid along the window.**

Past tense: the 10 Yr change in bp on a date, from the 20 trading days before it.

### Allowed names

fitted change in bp

walk-forward error

the last value carried forward

### Never called

a forecast or a prediction

tomorrow's move or a call

an outlook or a view on rates

# A filter sliding over time



20 daily changes of the 10 Yr before 2015-02-03; a filter of width 5 covers days 1 to 5, then 2 to 6, ...

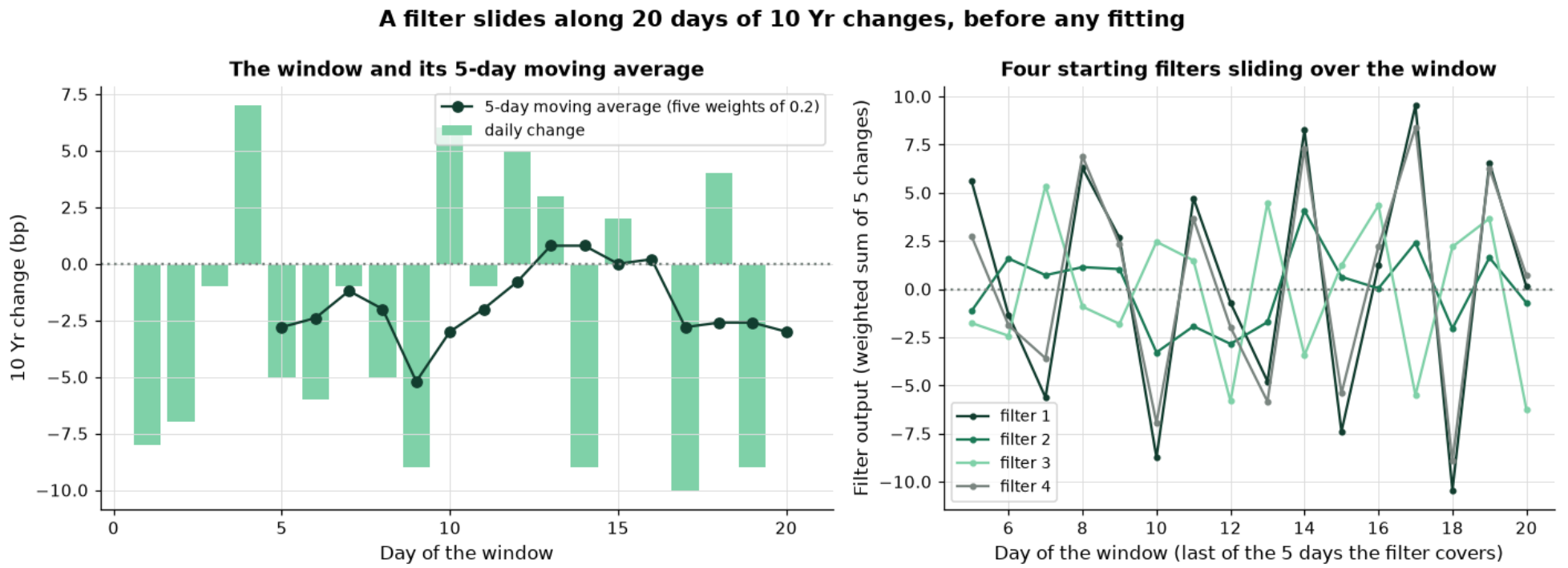
```
output[t] = sum of window[t + k] * weight[k] for k = 0 to 4, plus the bias
outputs: 20 - 5 + 1 = 16          kernel shape (width, channels, filters): (5, 1, 4)
```

### Five weights of 0.2 are a 5-day moving average

First output -2.8 bp; equal to pandas' rolling(5).mean() within 1e-12.

**A network fits its own weights: each filter is a moving sum with weights of its choosing.**

# The window and its moving average




A filter of equal weights smooths; four random filters give four different weighted moving sums. Nothing is fitted yet.

# A filter is a weighted moving sum

| | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|---|---|---|---|---|---|
| Change, bp | -8.0 | -7.0 | -1.0 | 7.0 | -5.0 |
| Weight | 0.351 | -0.4134 | 0.0233 | 0.4566 | -0.4714 |

**Products added, plus a bias of 0: 5.6165**

### Same five weights at every step

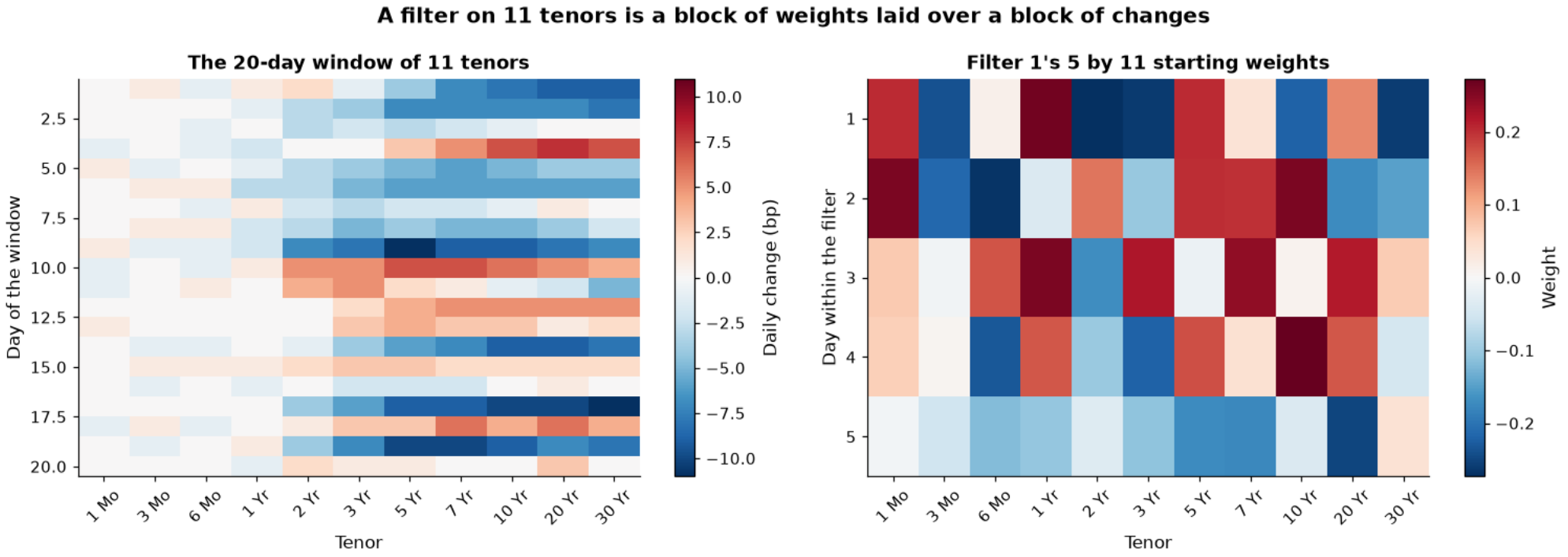
Day 11's dense network gave each day its own weight; a filter has 5 however long the window.

### 11 tenors as channels: kernel (5, 11, 4)

A filter is a block of 5 days by 11 tenors, 55 weights. Each output is one SUMPRODUCT over a rectangle; the output is still 16 steps per filter.

**Keras computes a cross-correlation: the kernel is not flipped.**

# A block of weights on a block of changes




**Each output multiplies a 5-day slice of the left grid, cell by cell, by the right grid and adds up: one SUMPRODUCT over a rectangle.**

# Pooling: one number per filter

**7 of 16**

filter 1's outputs set to 0 by ReLU

**2.8086**

average pooling, filter 1

**9.5347**

max pooling, filter 1

```
keras.layers.Conv1D(16, 5, activation="relu")   # (20, 1) -> (16, 16)
keras.layers.GlobalAveragePooling1D()           # (16, 16) -> (16,)
keras.layers.Dense(1)                            # (16,) -> 1, a change in either sign
```

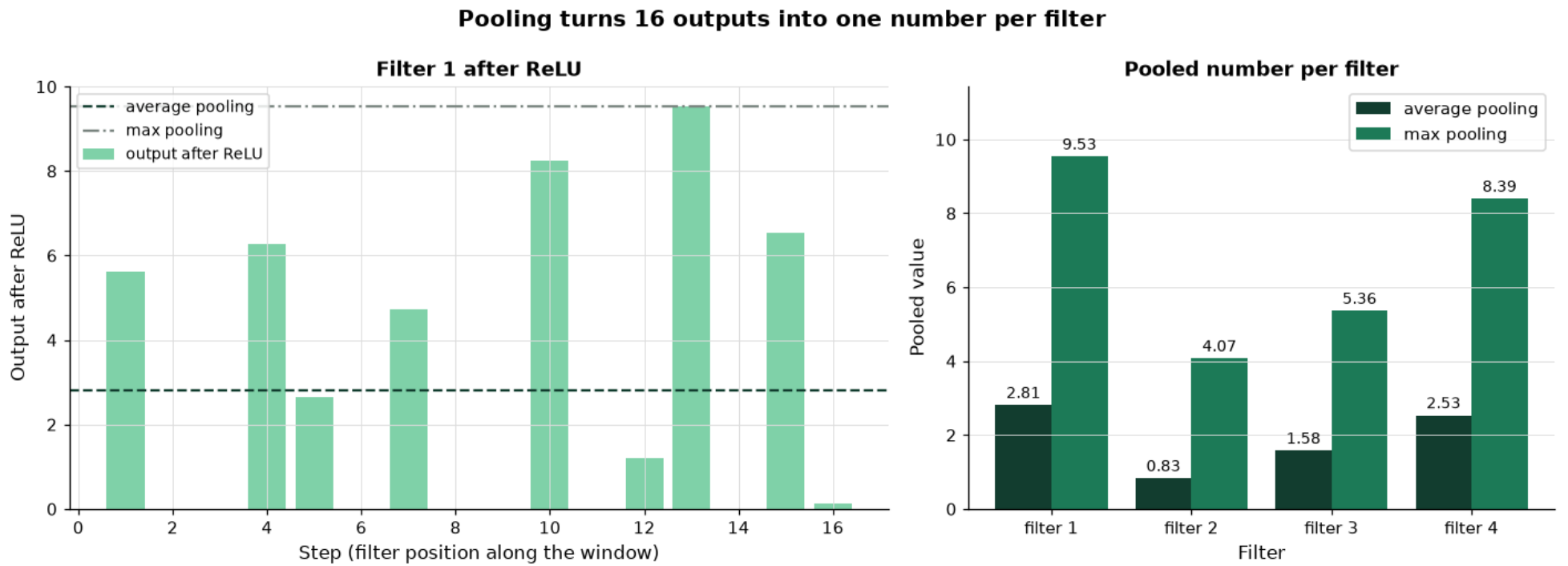
**113**

weights and biases, one tenor

**913**

weights and biases, 11 tenors

# Pooling, drawn




**ReLU set 7 of filter 1's 16 outputs to 0; pooling turns what is left into one number per filter.**

# Walk-forward: the Conv1D on the 10 Yr

| Model, 10 Yr alone | MAE, bp | Folds below zero change |
|---|---|---|
| zero change | 3.21 | - |
| training mean | 3.22 | - |
| dense, day 11 | 3.35 | 0 of 5 |
| SimpleRNN, day 12 | 3.31 | 0 of 5 |
| LSTM, day 12 | 3.23 | 0 of 5 |
| GRU, day 12 | 3.22 | 1 of 5 |
| Conv1D, today | 3.24 | 0 of 5 |

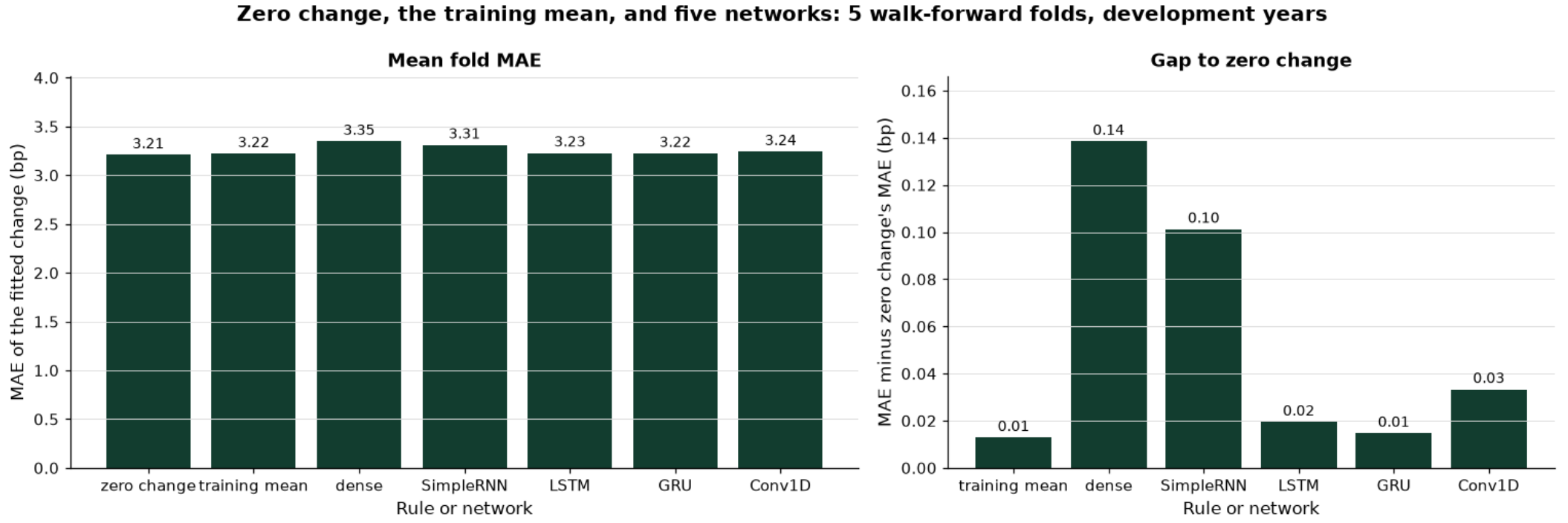
### Best epoch by fold, Conv1D

```
1, 50, 1, 1, 50
```

Fitted changes spread 0.25 bp against the actual 4.33 bp: little to fit, so it stays near the training mean.

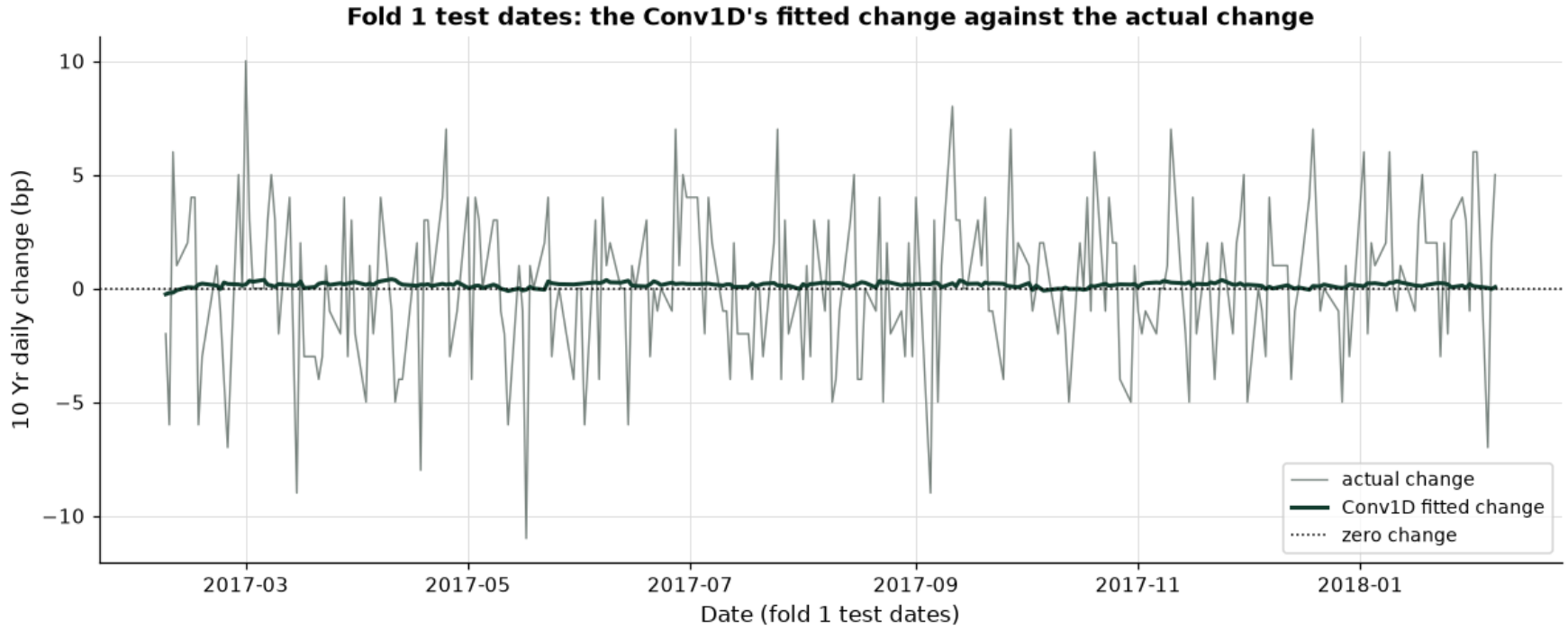
**Same five folds, same settings, reseeded every fold: days 11 and 12 rebuilt here.**

# Seven scores as bars




**Zero change, 3.21 bp, is the lowest bar; every other bar sits above it.**

# The Conv1D on fold one's dates




**The fitted changes spread 0.25 bp against the actual 4.33 bp: a network that finds little to fit stays near the training mean.**

# Many tenors as channels

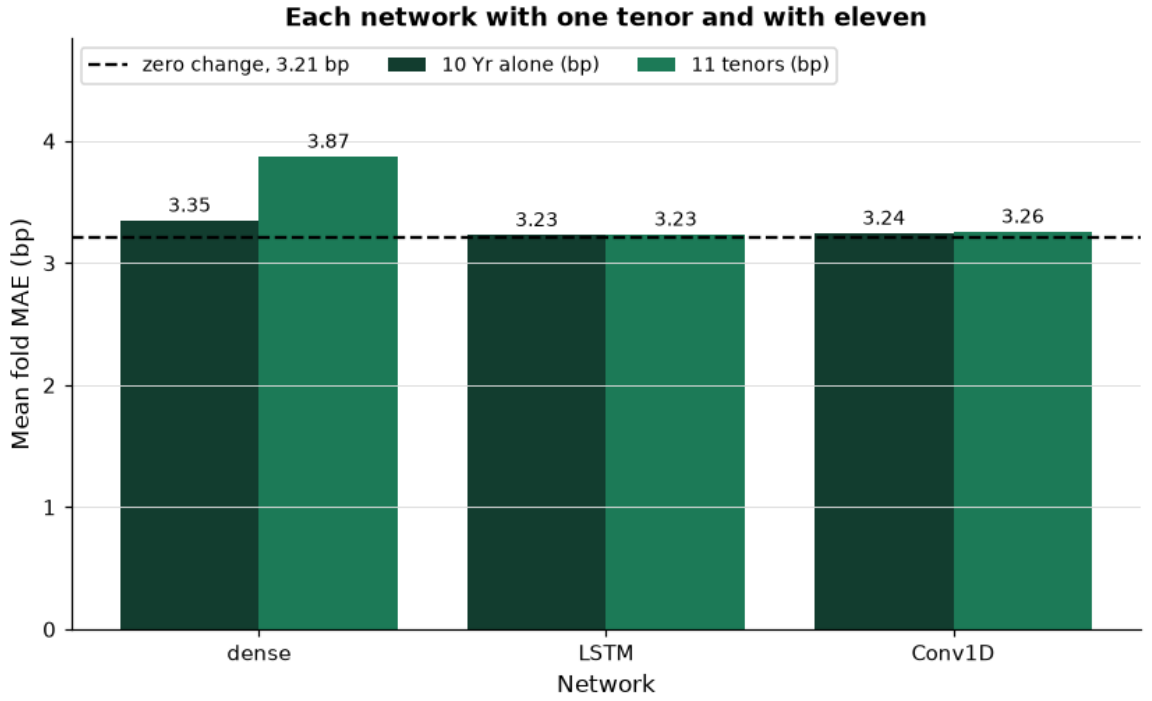
| Network | 10 Yr alone, bp | 11 tenors, bp | Folds below zero change |
|---|---|---|---|
| dense | 3.35 | 3.87 | 0 of 5 |
| LSTM | 3.23 | 3.23 | 1 of 5 |
| Conv1D | 3.24 | 3.26 | 0 of 5 |

Zero change on the same folds: 3.21 bp.

The 11-tenor Conv1D is not bit-identical run to run in its last digits (its MAE agreed to 6 decimals): printed as an MAE to 2 decimals and fold counts only.

**The flattened dense network got worse; the shared-weight networks barely moved.**

# One tenor or eleven, as bars




**The dense network's pair moves; the LSTM's and the Conv1D's are nearly equal. Every bar is above zero change.**

# Same-day changes stay out

| 10 Yr change against | Same date | One date earlier |
|---|---|---|
| 2 Yr change | 0.75 | 0.02 |
| 30 Yr change | 0.93 | -0.03 |

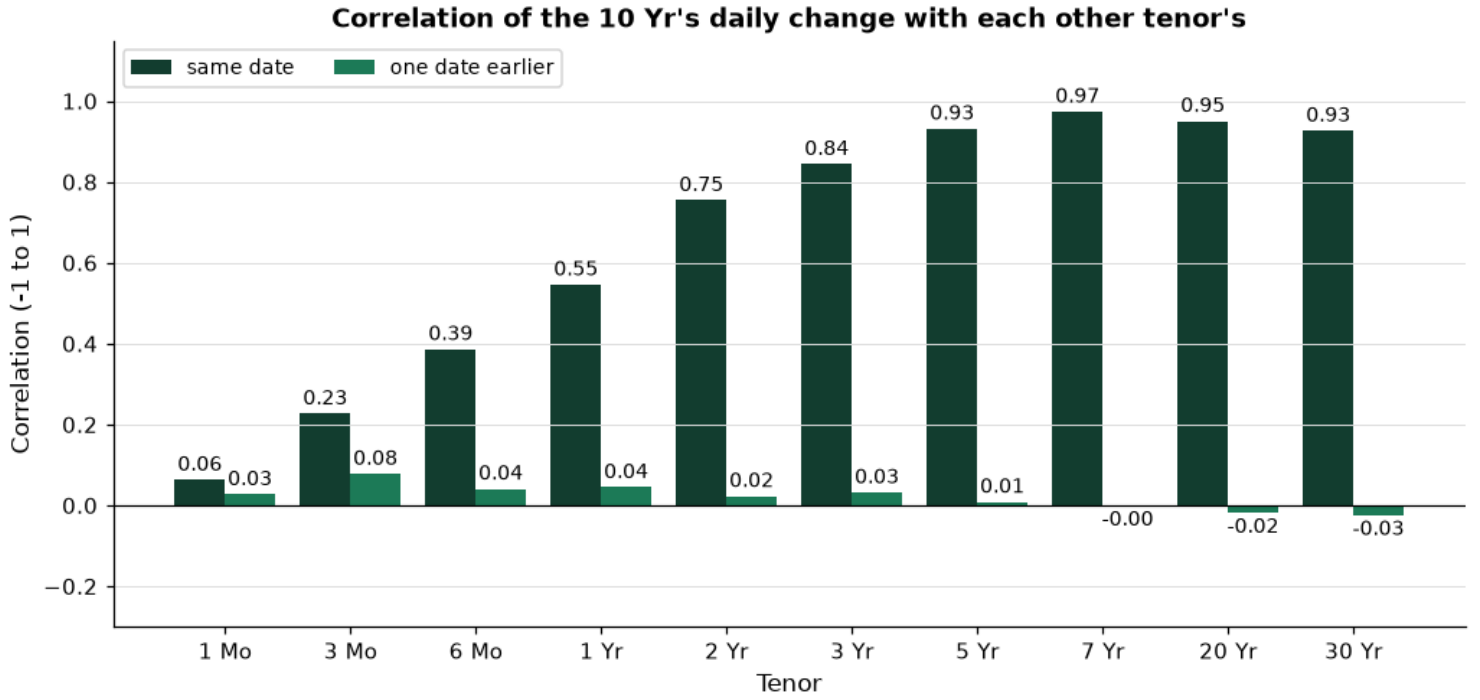
### Known only after the day

A window that ends on its target date holds other tenors' same-day changes.

```
netkit.assert_windows_past_only(changes.index[positions - 1], target_dates)   # passes
netkit.assert_windows_past_only(changes.index[positions], target_dates)
# AssertionError: 1980 of 1980 windows end on or after the date they are scored on
```

**make_windows: all 1,980 windows end the trading day before their target date.**

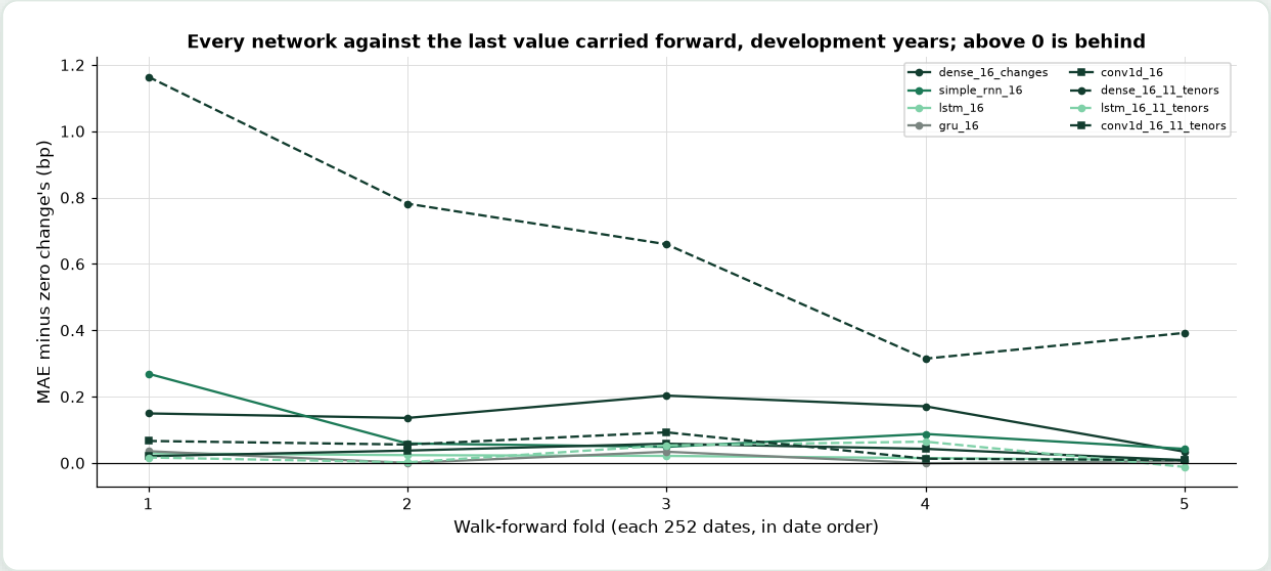
# Correlation, tenor by tenor




**Same-date bars rise from the shortest tenors to the longest; one-date-earlier bars sit near zero. One row too many in a window is the difference.**

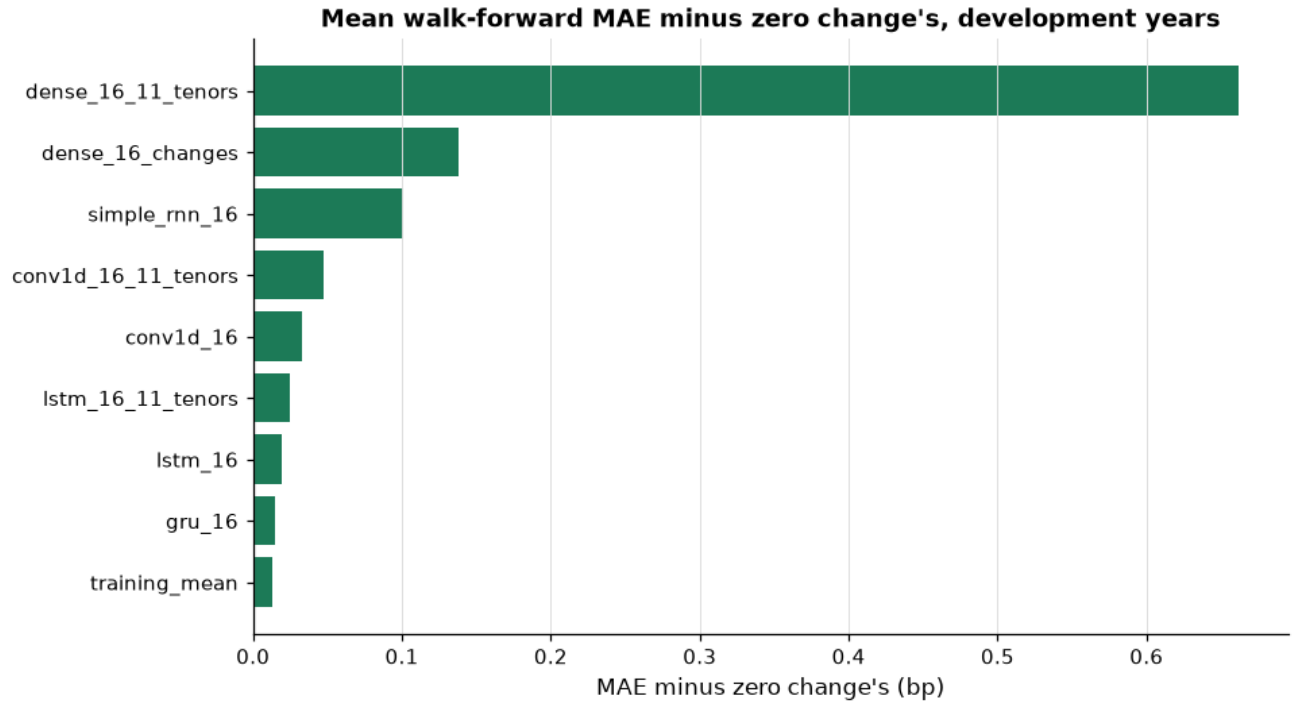
# Every sequence model so far

| Model | Ch. | MAE, bp | Below |
|---|---|---|---|
| zero_change | 1 | 3.21 | 0 |
| training_mean | 1 | 3.22 | 0 |
| dense_16_changes | 1 | 3.35 | 0 |
| simple_rnn_16 | 1 | 3.31 | 0 |
| lstm_16 | 1 | 3.23 | 0 |
| gru_16 | 1 | 3.22 | 1 |
| conv1d_16 | 1 | 3.24 | 0 |
| dense_16_11_tenors | 11 | 3.87 | 0 |
| lstm_16_11_tenors | 11 | 3.23 | 1 |
| conv1d_16_11_tenors | 11 | 3.26 | 0 |




**Zero change is the lowest line; no network beat it on average.**

# Gap to zero change, longest first




**The dense networks are the long bars; the LSTMs, the GRU, and the convolutions are short and close together. Every bar is above zero.**

# The candidates for day 14

**3.22 bp**

gru_16, 10 Yr alone: lowest mean MAE

**3.23 bp**

lstm_16, 10 Yr alone: second lowest

**3.21 bp**

zero change, beside them on day 14

### The rule

The two lowest mean walk-forward MAEs over the five development folds, on either input; every 11-tenor network was above its 10 Yr version. Chosen on development dates only; no held-out date fitted or scored.

**Chosen is not good: neither beats zero change, and the seed spread is day 14's.**

# Excel side by side

**Excel side by side (tested in Excel)**

| | |
|---|---|
| `=SUMPRODUCT(B2:B6,$K$2:$K$6)+$K$7` | C2 to C17: filter 1, first 5.6165 |
| `=SUMPRODUCT(B2:B6,$M$2:$M$6) = =AVERAGE(B2:B6)` | five weights of 0.2: a moving average |
| `=MAX(0,C2)` | ReLU, filled down |
| `=AVERAGE(D2:D17) and =MAX(D2:D17)` | pooling: 2.8086 and 9.5347 |
| `=SUMPRODUCT(B2:L6,$N$2:$X$6)+$N$7` | 11 tenors: a 5 by 11 rectangle |

**Equal to netkit within 1e-12 and to Keras within 1e-5. Excel cannot fit the weights, and has no walk-forward routine.**

# Save the day: the sequence log

### 1. Morning

Files as day 12 left them

### 2. Code

conv1d_by_hand, assert_windows_past_only

### 3. Log, rebuilt

Days 11 and 12, no levels row

### 4. Today's rows

Conv1D, and 11-tenor networks

```
!git -C "{work}" log --oneline --decorate
# (HEAD -> branch) marks the newest commit; day 14 adds a tag
```

**Columns: model, channels, window, rows, folds, mae_bp, zero_change_mae_bp, folds_below_zero_change. 56 tests pass.**

# The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

## Docstring

Which rows fit, which rows stop, which rows report: what you asked for.

→

2

## Code back

Written for the course in the style of an AI answer. Read it.

→

3

## Test first

Assert what the docstring promises. Run it: it fails.

→

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: 11-tenor windows end on the target date: fold 1 2.09 bp against zero change's 2.72 bp.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Compute a convolution by hand; check it against Keras
- Write a filter in Excel as SUMPRODUCT filled down
- Pool a filter's outputs: average and max
- Score a Conv1D walk-forward against zero change
- Keep same-day changes out of every window

**Week 3, Day 4** Case study: the held-out years, opened once